

No. of Functional Features Included in the Solution

Date	3 July 2026
TeamID	XXXXXX
ProjectName	SmartAgriculturalProductionOptimizationEngine
MaximumMarks	5Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	FeatureName	FeatureDescription	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Userlogin	Provides secure login for users and administrators.	Authentication	Done	10
2	SoilDataAnalysis	Analyzes soil parameters to support crop planning.	Soil analysis	Done	15
3	WeatherMonitoring	Retrieves weather information for better farming decisions	WeatherModule	Done	10
4	CropYield Prediction	Predicts crop production using Machine Learning algorithms.	Prediction engine	Done	20
5	Irrigation Optimization	Recommends optimal irrigation schedules based on crop and weather data	Optimization module	Done	15
6	Crop Recommendation	Suggests the most suitable crop based on soil and climate conditions.	Recommendation Module	Done	15
7	ReportGeneration	Generates reports with prediction results and recommendations.	ReportingModule	Done	10
8	AdminDashboard	Displays system information, analytics, and user management options.	Dashboard	Done	5

Feature Summary:

Metric	Count / Value
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TotalFeaturesPlanned	8
TotalFeaturesImplemented	8
Core/Must-HaveFeatures	6

Additional/Nice-to-HaveFeatures	2
FeaturesTested&Verified	8

FeatureCategoryBreakdown:

S.No	Category	FeaturesinCategory	ExampleFeatures
1	UserInterface(UI)	2	LoginPage,Dashboard
2	Backend/Logic	3	CropYieldPrediction,IrrigationOptimization,CropRecommendation
3	Database/Storage	1	FarmerDataandCropInformationStorage
4	API/ Integration	1	WeatherAPIIntegration
5	Security/Authentication	1	UserLoginAuthentication